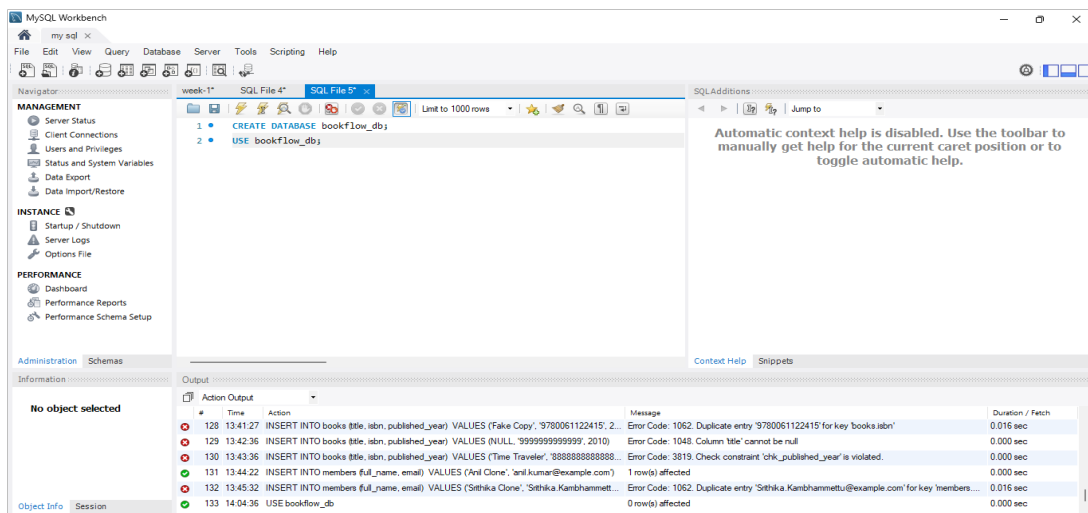


Sec:- 3

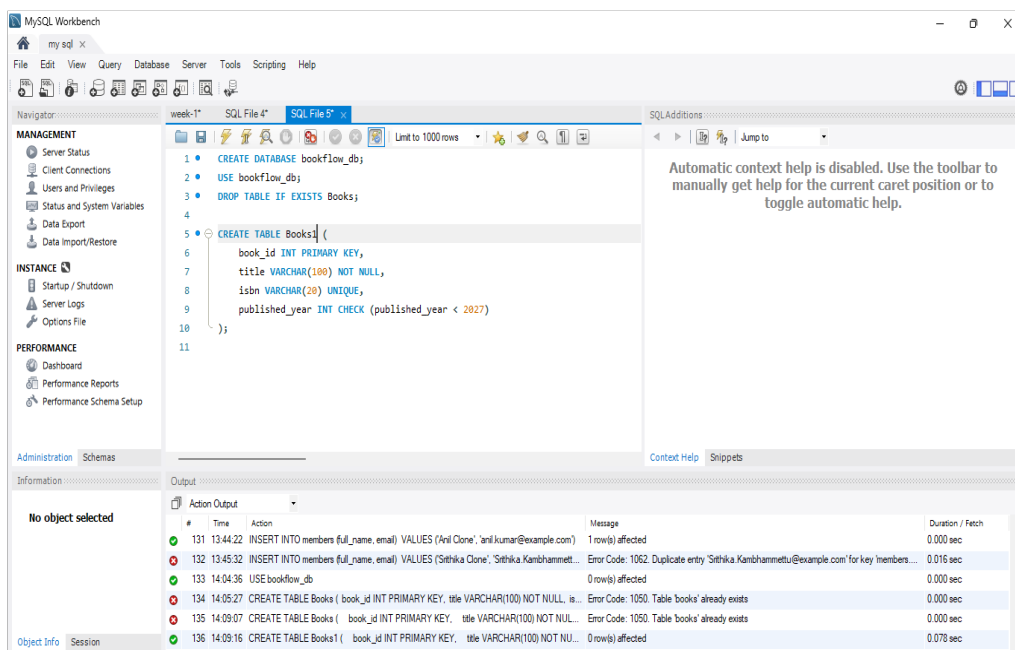
Roll no:- 2520030361

Week 2 Practical

step-1:



Step 2-



Step 3 & 4:

MySQL Workbench interface showing the SQL editor and the Output pane. The SQL editor contains the following query:

```
10 ;
11 INSERT INTO Books (book_id, title, isbn, published_year) VALUES
12 (1, 'The Great Gatsby', '9780743273565', 1925),
13 (2, 'To Kill a Mockingbird', '9780061120084', 1960),
14 (3, '1984', '9780451524935', 1949);
15 SELECT * FROM Books;
16
```

The Output pane shows the results of the queries:

#	Time	Action	Message	Duration / Fetch
132	13:45:32	INSERT INTO members (full_name, email) VALUES ('Sithika Clone', 'Sithika.Kambhammettu@example.com')	Error Code: 1062. Duplicate entry 'Sithika.Kambhammettu@example.com' for key 'members....	0.016 sec
133	14:04:36	USE bookflow_db	0 row(s) affected	0.000 sec
134	14:05:27	CREATE TABLE Books (book_id INT PRIMARY KEY, title VARCHAR(100) NOT NULL, is...	Error Code: 1050. Table 'books' already exists	0.000 sec
135	14:09:07	CREATE TABLE Books (book_id INT PRIMARY KEY, title VARCHAR(100) NOT NUL...	Error Code: 1050. Table 'books' already exists	0.000 sec
136	14:09:16	CREATE TABLE Books1 (book_id INT PRIMARY KEY, title VARCHAR(100) NOT NUL...	0 row(s) affected	0.078 sec
137	14:11:09	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec

Step 5,6&7:

MySQL Workbench interface showing the SQL editor and the Output pane. The SQL editor contains the following queries:

```
1 CREATE DATABASE IF NOT EXISTS bookflow_db;
2 USE bookflow_db;
3 DROP TABLE IF EXISTS Books;
4 CREATE TABLE Books (
5   book_id INT PRIMARY KEY,
6   title VARCHAR(100) NOT NULL,
```

The Output pane shows the results of the queries:

#	Time	Action	Message	Duration / Fetch
180	09:25:36	INSERT INTO Books (book_id, title, isbn, published_year) VALUES (1, 'The Great Gatsby', ...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec
181	09:25:36	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
182	09:25:36	DROP TABLE IF EXISTS Members	0 row(s) affected	0.016 sec
183	09:25:36	CREATE TABLE Members (member_id INT PRIMARY KEY, full_name VARCHAR(100), e...	0 row(s) affected	0.031 sec
184	09:25:36	INSERT INTO Members (member_id, full_name, email) VALUES (101, 'John Smith', 'john.smi...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec
185	09:25:36	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned	0.016 sec / 0.000 sec

Step 8:

The screenshot shows the MySQL Workbench interface. The SQL Editor contains the following queries:

```
20 );
21 INSERT INTO Members (member_id, full_name, email) VALUES
22 (101, 'John Smith', 'john.smith@email.com'),
23 (102, 'Emma Wilson', 'emma.wilson@email.com'),
24 (103, 'Michael Brown', 'michael.brown@email.com');
25 SELECT * FROM Members;
26 CREATE TABLE Loans (
27 loan_id INT PRIMARY KEY,
28 member_id INT,
29 book_id INT,
30 loan_date DATE,
31 FOREIGN KEY (member_id) REFERENCES Members(member_id),
32 FOREIGN KEY (book_id) REFERENCES Books(book_id)
33 );
34
```

The Output pane shows the results of the queries:

#	Time	Action	Message	Duration / Fetch
181	09:25:36	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
182	09:25:36	DROP TABLE IF EXISTS Members	0 row(s) affected	0.016 sec
183	09:25:36	CREATE TABLE Members (member_id INT PRIMARY KEY, full_name VARCHAR(100), e...	0 row(s) affected	0.031 sec
184	09:25:36	INSERT INTO Members (member_id, full_name, email) VALUES (101, 'John Smith', 'John.smi...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec
185	09:25:36	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned	0.016 sec / 0.000 sec
186	09:29:32	CREATE TABLE Loans (loan_id INT PRIMARY KEY, member_id INT, book_id INT, loan...	0 row(s) affected	0.047 sec

Step 9&10:

The screenshot shows the MySQL Workbench interface. The SQL Editor contains the following queries:

```
35 (1, 101, 1, '2025-01-05'), (2, 102, 2, '2025-01-06'),
36 (3, 103, 3, '2025-01-10'), (4, 101, 2, '2025-02-01'),
37 (5, 102, 1, '2025-02-05'), (6, 103, 2, '2025-02-12'),
38 (7, 101, 3, '2025-03-01'), (8, 102, 3, '2025-03-07'),
39 (9, 103, 1, '2025-03-15'), (10, 101, 1, '2025-04-01');
40 SELECT * FROM Loans;
```

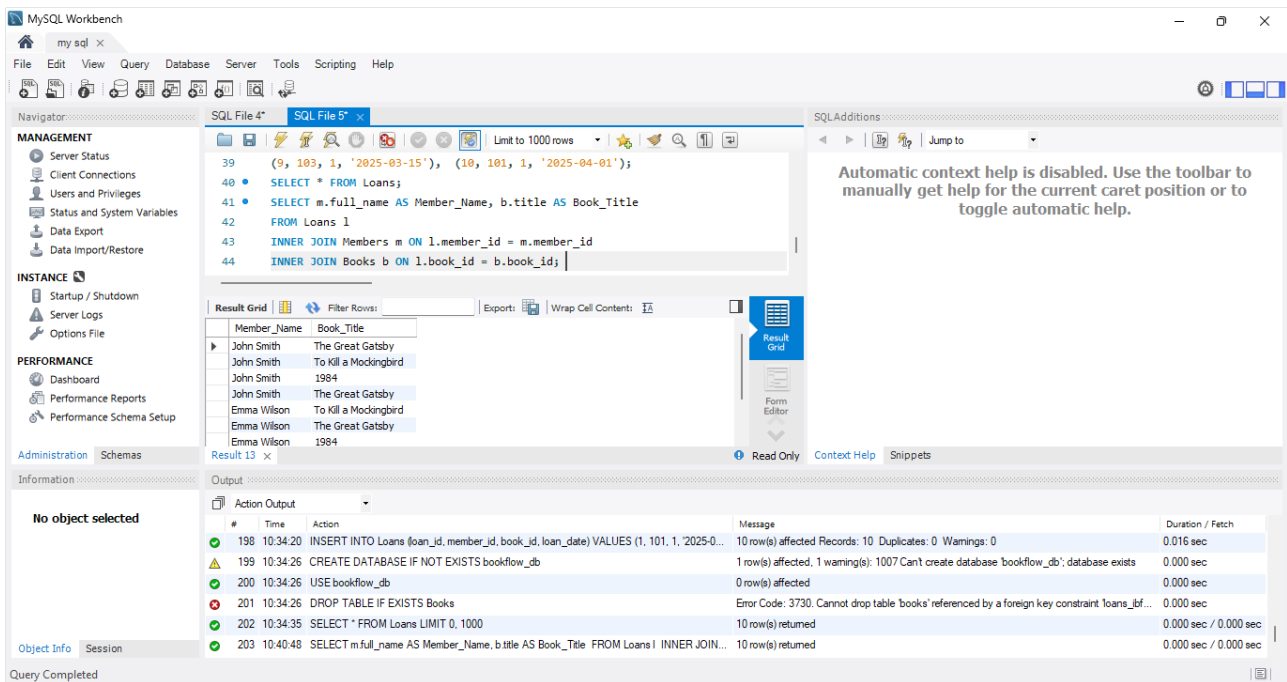
The Result Grid shows the data for the 'Loans' table:

loan_id	member_id	book_id	loan_date
1	101	1	2025-01-05
2	102	2	2025-01-08
3	103	3	2025-01-10
4	101	2	2025-02-01
5	102	1	2025-02-05
6	103	2	2025-02-12
7	101	3	2025-03-01

The Output pane shows the results of the queries:

#	Time	Action	Message	Duration / Fetch
197	10:32:49	CREATE TABLE Loans (loan_id INT PRIMARY KEY, member_id INT, book_id INT, loan...	Error Code: 1050. Table 'loans' already exists	0.000 sec
198	10:34:20	INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES (1, 101, 1, '2025-0...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.016 sec
199	10:34:26	CREATE DATABASE IF NOT EXISTS bookflow_db	1 row(s) affected, 1 warning(s): 1007 Can't create database 'bookflow_db'; database exists	0.000 sec
200	10:34:26	USE bookflow_db	0 row(s) affected	0.000 sec
201	10:34:26	DROP TABLE IF EXISTS Books	Error Code: 3730. Cannot drop table 'books' referenced by a foreign key constraint 'loans_bf...	0.000 sec
202	10:34:35	SELECT * FROM Loans LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec

Step 11:



MySQL Workbench

my sql x

File Edit View Query Database Server Tools Scripting Help

Navigator

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

Administration Schemas

Information

No object selected

Object Info Session

SQL File 4* SQL File 5*

Limit to 1000 rows

SQL

```
39 (9, 103, 1, '2025-03-15'), (10, 101, 1, '2025-04-01');
40 SELECT * FROM Loans;
41 SELECT m.full_name AS Member_Name, b.title AS Book_Title
42 FROM Loans l
43 INNER JOIN Members m ON l.member_id = m.member_id
44 INNER JOIN Books b ON l.book_id = b.book_id;
```

Result Grid

Member_Name	Book_Title
John Smith	The Great Gatsby
John Smith	To Kill a Mockingbird
John Smith	1984
John Smith	The Great Gatsby
Emma Wilson	To Kill a Mockingbird
Emma Wilson	The Great Gatsby
Emma Wilson	1984

Result 13 x

Read Only Context Help Snippets

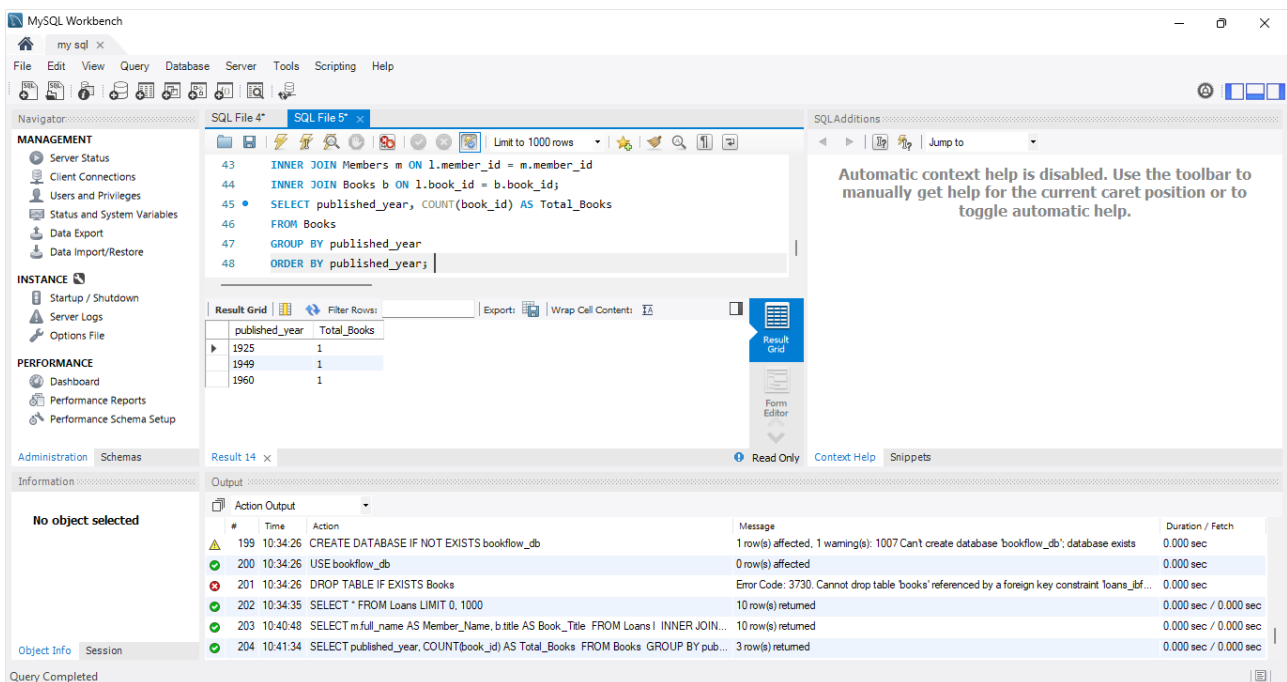
Output

Action Output

#	Time	Action	Message	Duration / Fetch
198	10:34:20	INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES (1, 101, 1, '2025-0...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.016 sec
199	10:34:26	CREATE DATABASE IF NOT EXISTS bookflow_db	1 row(s) affected, 1 warning(s): 1007 Can't create database 'bookflow_db'; database exists	0.000 sec
200	10:34:26	USE bookflow_db	0 row(s) affected	0.000 sec
201	10:34:26	DROP TABLE IF EXISTS Books	Error Code: 3730. Cannot drop table 'books' referenced by a foreign key constraint 'loans_bf...	0.000 sec
202	10:34:35	SELECT * FROM Loans LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
203	10:40:48	SELECT m.full_name AS Member_Name, b.title AS Book_Title FROM Loans l INNER JOIN...	10 row(s) returned	0.000 sec / 0.000 sec

Query Completed

Step 12:



MySQL Workbench

my sql x

File Edit View Query Database Server Tools Scripting Help

Navigator

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

Administration Schemas

Information

No object selected

Object Info Session

SQL File 4* SQL File 5*

Limit to 1000 rows

SQL

```
43 INNER JOIN Members m ON l.member_id = m.member_id
44 INNER JOIN Books b ON l.book_id = b.book_id;
45 SELECT published_year, COUNT(book_id) AS Total_Books
46 FROM Books
47 GROUP BY published_year
48 ORDER BY published_year;
```

Result Grid

published_year	Total_Books
1925	1
1949	1
1960	1

Result 14 x

Read Only Context Help Snippets

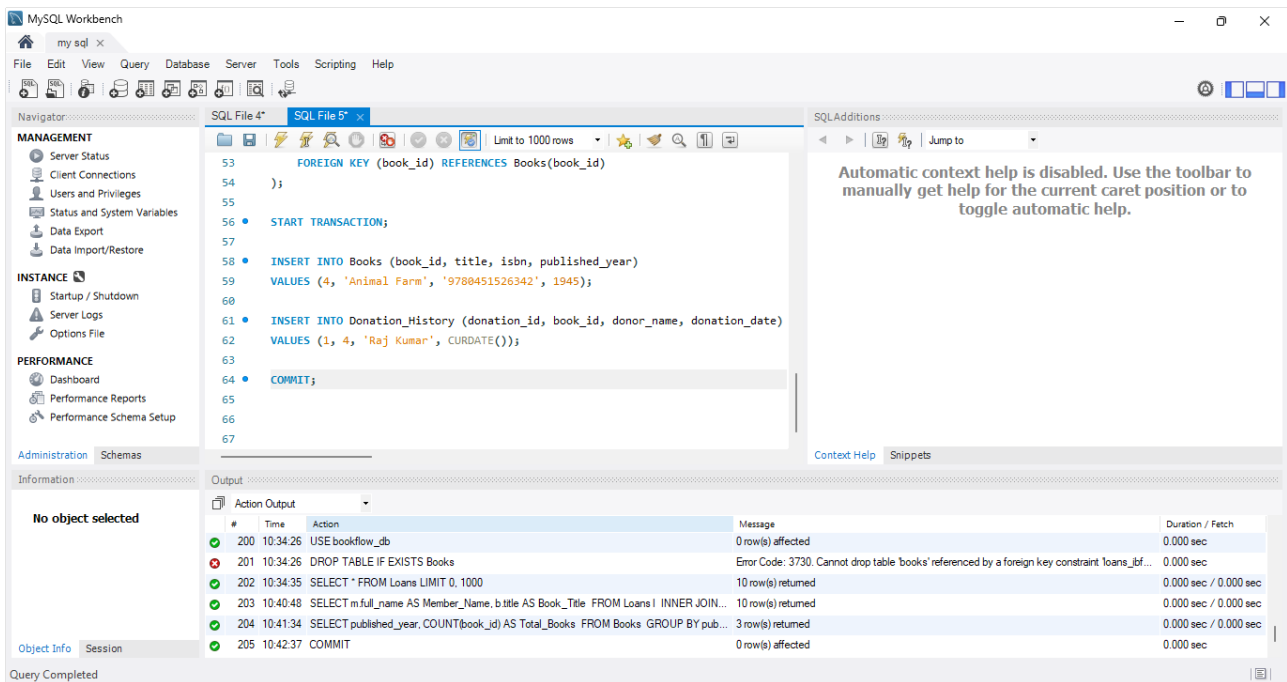
Output

Action Output

#	Time	Action	Message	Duration / Fetch
199	10:34:26	CREATE DATABASE IF NOT EXISTS bookflow_db	1 row(s) affected, 1 warning(s): 1007 Can't create database 'bookflow_db'; database exists	0.000 sec
200	10:34:26	USE bookflow_db	0 row(s) affected	0.000 sec
201	10:34:26	DROP TABLE IF EXISTS Books	Error Code: 3730. Cannot drop table 'books' referenced by a foreign key constraint 'loans_bf...	0.000 sec
202	10:34:35	SELECT * FROM Loans LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
203	10:40:48	SELECT m.full_name AS Member_Name, b.title AS Book_Title FROM Loans l INNER JOIN...	10 row(s) returned	0.000 sec / 0.000 sec
204	10:41:34	SELECT published_year, COUNT(book_id) AS Total_Books FROM Books GROUP BY pub...	3 row(s) returned	0.000 sec / 0.000 sec

Query Completed

Step 13-16:



MySQL Workbench

my sql x

File Edit View Query Database Server Tools Scripting Help

SQL File 5* x

Limit to 1000 rows

```
53 FOREIGN KEY (book_id) REFERENCES Books(book_id)
54 );
55
56 • START TRANSACTION;
57
58 • INSERT INTO Books (book_id, title, isbn, published_year)
59 VALUES (4, 'Animal Farm', '9780451526342', 1945);
60
61 • INSERT INTO Donation_History (donation_id, book_id, donor_name, donation_date)
62 VALUES (1, 4, 'Raj Kumar', CURDATE());
63
64 • COMMIT;
65
66
67
```

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

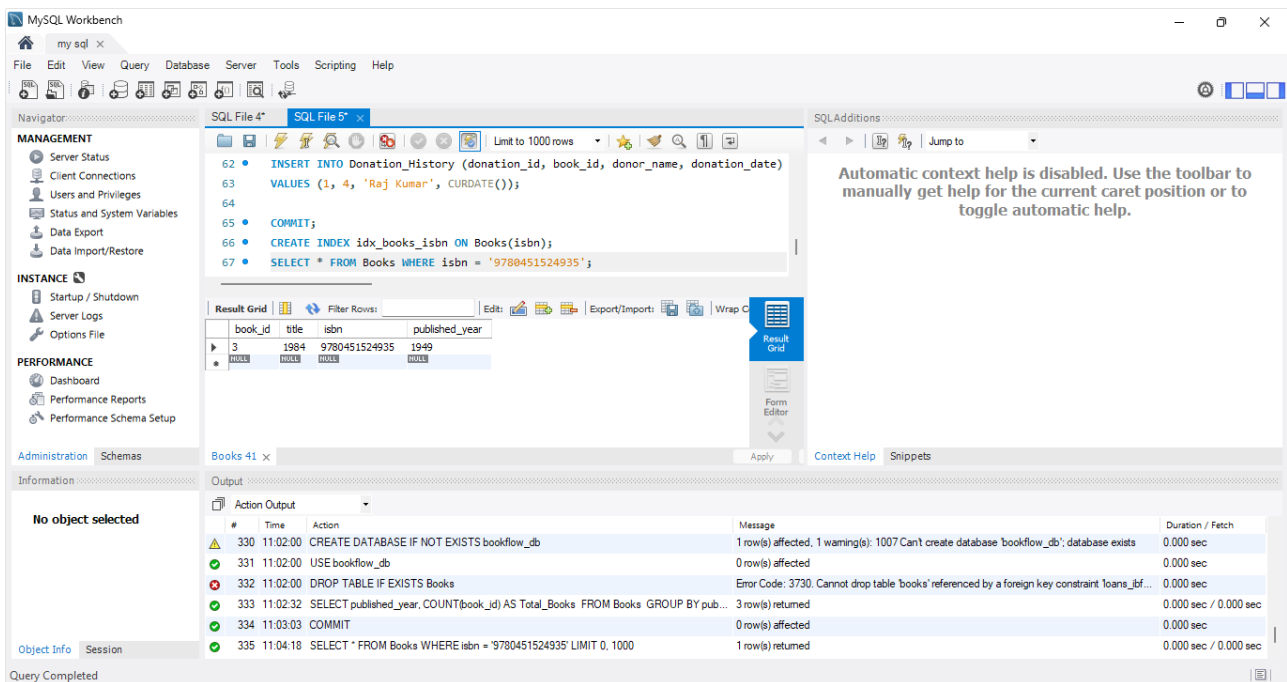
Output

Action Output

#	Time	Action	Message	Duration / Fetch
200	10:34:26	USE bookflow_db	0 row(s) affected	0.000 sec
201	10:34:26	DROP TABLE IF EXISTS Books	Error Code: 3730. Cannot drop table 'books' referenced by a foreign key constraint 'loans_bf...	0.000 sec
202	10:34:35	SELECT * FROM Loans LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
203	10:40:48	SELECT m.full_name AS Member_Name, b.title AS Book_Title FROM Loans l INNER JOIN...	10 row(s) returned	0.000 sec / 0.000 sec
204	10:41:34	SELECT published_year, COUNT(book_id) AS Total_Books FROM Books GROUP BY pub...	3 row(s) returned	0.000 sec / 0.000 sec
205	10:42:37	COMMIT	0 row(s) affected	0.000 sec

Query Completed

Step 17&18:



MySQL Workbench

my sql x

File Edit View Query Database Server Tools Scripting Help

SQL File 5* x

Limit to 1000 rows

```
62 • INSERT INTO Donation_History (donation_id, book_id, donor_name, donation_date)
63 VALUES (1, 4, 'Raj Kumar', CURDATE());
64
65 • COMMIT;
66 • CREATE INDEX idx_books_isbn ON Books(isbn);
67 • SELECT * FROM Books WHERE isbn = '9780451524935';
```

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Result Grid

book_id	title	isbn	published_year
3	1984	9780451524935	1949

Books 41 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
330	11:02:00	CREATE DATABASE IF NOT EXISTS bookflow_db	1 row(s) affected, 1 warning(s): 1007 Can't create database 'bookflow_db': database exists	0.000 sec
331	11:02:00	USE bookflow_db	0 row(s) affected	0.000 sec
332	11:02:00	DROP TABLE IF EXISTS Books	Error Code: 3730. Cannot drop table 'books' referenced by a foreign key constraint 'loans_bf...	0.000 sec
333	11:02:32	SELECT published_year, COUNT(book_id) AS Total_Books FROM Books GROUP BY pub...	3 row(s) returned	0.000 sec / 0.000 sec
334	11:03:03	COMMIT	0 row(s) affected	0.000 sec
335	11:04:18	SELECT * FROM Books WHERE isbn = '9780451524935' LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec

Query Completed